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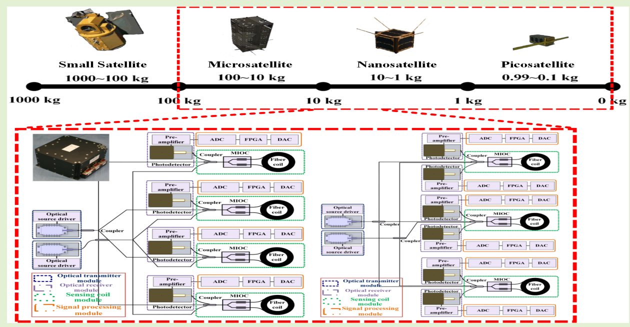
Configuration Optimization in Miniature Interferometric Fiber-Optic Gyroscopes for Space Application

Kun Ma, Ningfang Song, Jing Jin, Jiliang He, and Enrico Zio

***Abstract*—With the development of small satellites, the space-borne interferometric fiber-optic gyroscopes (IFOGs) need to satisfy the requirements of high reliability, light weight and small size. Traditional multiple-axis redundant configurations are no longer applicable for miniature IFOGs. In this paper, a novel configuration reliability optimization method is presented for IFOGs, whereas limiting weight and volume. Firstly, a fault tree analysis for the space-borne single-axis IFOG is carried out and the effects of the space environment on IFOG components are discussed. Then, based on the failure features and performance degradation characteristics of the IFOG components, the reliability model is deduced. The configuration reliability optimization method is proposed for the first time to achieve high reliability of space-borne IFOGs, without increasing weight and volume. Traditional four single-axis IFOG configuration is employed to verify the effectiveness of the proposed method, the optimized results show that with our experimental IFOG parameters, the reliability is improved by 2.3%-3.1% in the first year, 21.4%-22.1% in the third year and 53.9%-55.0% in the fifth year, whereas the total weight dropped by 8.3% and the total volume is almost same. Therefore, the proposed method is efficient and easy to implement, which actually has been used in prototypes and products.**



***Index Terms*—Interferometric fiber-optic gyroscopes, reliability, configuration optimization.**

I. Introduction

INTERFEROMETRIC fiber-optic gyroscopes (IFOGs) measure the inertial rotation rate around a fixed-axis by explaining the Sagnac effect and exhibit many advantages, including fully solid-state configuration, high performance, high reliability and high dynamic range. They have been widely used in various space applications, especially for satellites [1]. However, due to the long-term exposure in the space environment, the components of the space-borne IFOGs will be affected by extreme temperature and ionizing radiation during their mission, which would cause performance degradation, reliability loss and possible even the failure of the IFOGs.

Redundance is a typical and useful method to guarantee the high reliability of the space-borne IFOGs. Usually, a three-dimensional attitude is required to realize satellite attitude control, which means at least three orthogonal axes of the space-borne IFOG system must be effective during the space mission [2]. A single-axis closed-loop IFOG is a series system, and once one of its components fails, the whole single-axis IFOG will fail. Correspondingly, preliminary studies about the redundant configurations are mainly including quad redundant configurations and hexad redundant configurations [3], [4]. The quad redundant configurations consist of the pyramidal structure and the orthogonal structure, whereas hexad redundant configurations consist of the hexahedral structure and the dodecahedral structure. All these configurations can achieve

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fault-tolerance and high-reliability by adding redundant single-axis IFOGs on the basis of the three-axis IFOG configuration. Unfortunately, they also heavily increase the weight and the volume of the whole IFOG system. Currently, the requirements of light weight, small size and low cost have become dominant in the design of satellites, making them accessible to the research, educational and commercial sectors [5]. Therefore, the traditional redundance method is no longer suitable to the miniature space-borne IFOGs and the efforts are underway to achieve miniature IFOGs in a high-reliability design for space application.

Enhancing the anti-environment disturbance ability of the IFOGs' critical components has been theoretically proved to be an effective way to deal with the problem discussed before. For optical components, Wei *et al.* proposed an optimized small-diameter polarization maintaining photonic crystal fiber (PM-PCF), which has good thermal characteristics, superior radiation resistance, and it is suitable for space-borne miniature FOGs [6], [7]. Roberts *et al.* developed a photonic-bandgap fiber (PBF) to replace conventional PMFs, with several benefits for space use [8]. However, the back reflection and backscattering in PCFs and PBFs will cause an increase of fiber loss and a decrease of IFOG accuracy, which is still a problem demanding prompt solution [9], [10]. More importantly, those reported new-type fibers are all based on complicated processing techniques. For electronic components, high quality and low cost commercial off-the-shelf components have been used to replace the MIL quality components for space application, in order to improve the performance and reduce the cost of low orbit satellites and the corresponding equipment [11], [12]. However, the performance of those electronic components needs to be verified by sending miniature scientific radiation monitoring payloads or small satellites to space, or by ground-based radiation tests.

In order to solve this problem, in this work, we, for the first time to the best of our knowledge, demonstrate a simple, novel configuration reliability optimization method for miniature IFOGs and provide the reliability optimal IFOG configurations in the limited range of the weight and the volume. Compared to the existing methods, the established method improves the reliability of miniature space-borne IFOGs without increasing the weight and decreasing the precision by multiplexing the robust parts and designing the vulnerable parts with redundance. To assess the reliability of the space-borne IFOGs, a fault tree analysis (FTA) is firstly carried out, considering the characteristics of the IFOG components in the space environment. For obtaining the optimal configurations, a reliability optimization algorithm is developed. With this algorithm, the reliability of all the configurations satisfying the IFOG working principles and the requirements of the weight and the volume is calculated and compared.

The overall organization of the paper is briefly summarized as follows. First of all, the reliability of the space-borne single-axis IFOG configuration and the multiple-axis redundant IFOG configurations are modeled in Section II. Then, a FTA for the space-borne single-axis IFOG is carried out and the effects of the space environment on the IFOG components are analyzed. Based on the analysis results, the reliability

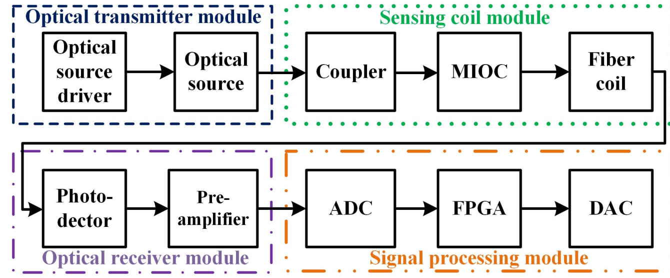

Fig. 1. Reliability block diagram of a typical single-axis closed-loop IFOG.

distribution function of the IFOG components is established in Section III. Next, a configuration reliability optimization method and the corresponding algorithm are presented in Section IV. To demonstrate this method, we optimize the traditional four-axis redundant IFOG configuration and give a comparison of the optimized results. After that, the validation based on the Monte Carlo method (MCM), the results of the ground-based tests and in-orbit tests of the products using the proposed optimized configurations are provided in Section V. Finally, conclusions are drawn in Section VI.

II. Fault Tree Analysis of Space-Borne IFOGs

For the analysis of different IFOG configurations, a reliability block diagram method is used to model the reliability of the typical single-axis closed-loop IFOG, in which the IFOG is divided into 4 basic blocks based on their functions and characteristics, ie., optical transmitter module, optical receiver module, sensing coil module and signal processing module, as shown in Fig. 1. The optical transmitter module and the optical receiver module are composed of optoelectronic components and active optical components. The former includes an optical source driver and an optical source, whereas the latter involves a photodetector and a pre-amplifier. The sensing coil module contains passive optical components, ie., a coupler, a multifunction integrated-optical circuit (MIOC) and a fiber coil. The signal processing module consists of electronic components, ie., an analog/digital converter (ADC), a FPGA and a digital/analog converter (DAC).

On this foundation, a FTA is used to analyze the effects of the space environment on the IFOG components, as given in Fig. 2. It is assumed that the failure of a space-borne single-axis IFOG is the top event (TE), which is built up through top-down deduction. The failure of the IFOG components is taken as the intermediate events (IEs), whereas the failure modes of each component are considered basic events (BEs). All the failure events are listed in TABLE I. As shown in Fig. 2, the failure of the space-borne single-axis IFOG has 28 IEs numbered *G1, G2,…, G28* and 9 BEs numbered *G29,…, G37*, all leading to the TE which is represented as T. The fault tree is a combination of 'OR' gates, so the reliability of a single-axis IFOG R can be expressed as:

$$R(T) = R_{G1} \cdot R_{G2} \cdot R_{G3} \cdot R_{G4} \tag{1}$$

where R_{Gi} is the reliability of the event G_i, $i = 1, 2..., 37$, and it can be estimated using Boolean logic and possibility theory [13].

TABLE I
FAILURE EVENTS OF THE SPACE-BORNE SINGLE-AXIS IFOG

Sym-bol	Event	Ty-pe	Sym-bol	Event	Ty-pe	Sym-bol	Event	Ty-pe
T	Space-borne single-axis IFOG failure	TE	G13	FPGA failure	IE	G26	Temperature control circuit failure	IE
G_1	Sensing coil module failure	IE	G14	DAC failure	IE	G27	Photodiode failure	IE
G2	Optical transmitter module failure	IE	G15	Power supply circuit failure	IE	G28	Fiber failure	IE
G3	Optical receiver module failure	IE	G16	communication circuit failure	IE	G29	Trans-Impedance Amplifier failure	BE
G4	Signal processing module failure	IE	G17	Fiber failure	IE	G30	Radiation total dose damage effects in devices	BE
G5	Coupler failure	IE	G18	Adhesives failure	IE	G31	Aging-induced performance	BE
G6	MIOC failure	IE	G19	Electrodes failure	IE		degradation	BE
G7	Fiber coil failure	IE	G20	Fiber failure	IE	G32	Single event effects	BE
G8	Optical source failure	IE	G21	Waveguide failure	IE	G33	Stress-induced fiber damage	BE
G9	Optical source driver failure	IE	G22	Fiber failure	IE	G34	Radiation induced attennuation	BE
G10	Photodetector failure	IE	G23	Optical source tube core failure	IE	G35	Electrode short circuit	BE
G11	Pre-amplifier failure	IE	G24	Fiber failure	IE	G36	Electrode breakdown	BE
G12	ADC failure	IE	G25	Constant current circuit failure	IE	G37	Electrode movement	BE

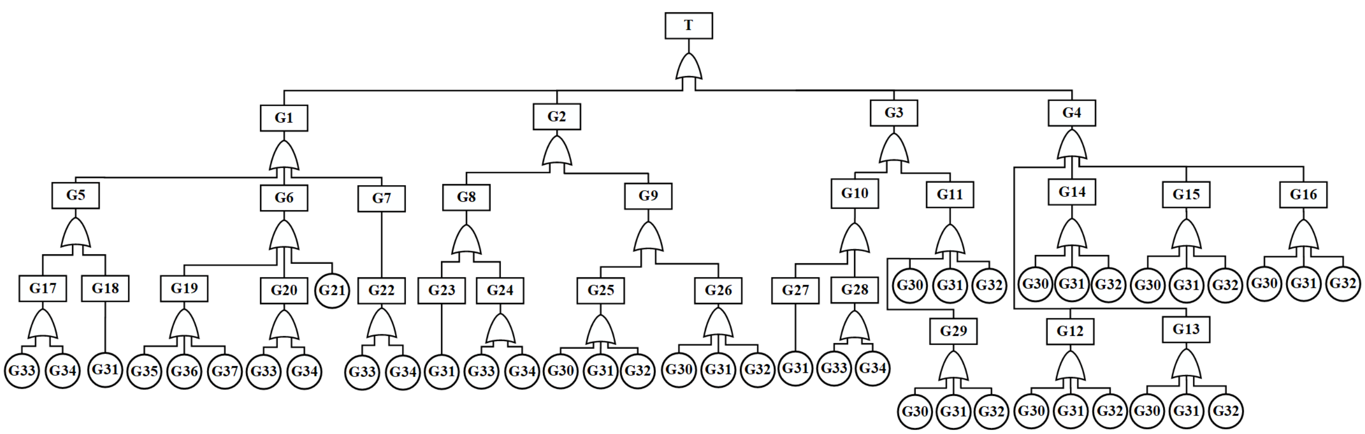

Fig. 2. FTA of the space-borne single-axis IFOG.

III. RELIABILITY CALCULATION OF IFOG CONFIGURATIONS

A. Reliability of IFOG Components

The FTA of Fig. 2 shows that different kinds of IFOG components are affected by different environmental factors in space. The optical components, especially the fiber coils, can be affected by radiation-induced attenuation (RIA). The closed-loop distributed IFOG tests have demonstrated the very little effect of proton irradiation on the MIOC and the coupler [14]. As for optoelectronic components in space-borne IFOGs, they are mainly influenced by temperature. Compared with the fiber coil, the fiber length in the optical source and the photodetector are very short. The effect of irradiation on these components is negligible. Besides, the radiation resistance of the light-emitting diode optical source and the InGaAs-pin-type photodetector was also verified by experiments [15]. The experiment results demonstrate that the degradation of the super-luminescent diode (SLD) optical source and the PIN-FET photodetectors merely depend on temperature and working time, rather than on radiation dose rate or cumulated dose. The electronic components in IFOGs can be affected by the temperature and the radiation of the space environment. Actually, the aero-space and military-grade electronic components usually have radiation resistance to meet the needs of space missions. For a high-reliability component, it will not be abraded or eroded like mechanical equipment, and its failure probability does not increase with time and obeys the exponential distribution [16]. Thus, we use the exponential distribution to describe the reliability of the IFOG components (except the fiber coil) with the corresponding failure rates summarized in TABLE II.

TABLE II
FAILURE RATES OF COMPONENTS IN IFOGs

IFOG Components		Failures /per Million Hrs
Optical transmitter module	Optical source	6.64
	Optical source driver	0.01
Optical receiver module	PIN-FET	3.08
	pre-amplifier	0.01
Sensing coil module	Coupler	0.25
	MIOC	0.056
Signal processing module	ADC	0.01
	FPGA	0.00174
	DAC	0.01

The typical optical components and optoelectronic components used in IFOGs are produced by Beijing SWT Science & Technology Development Co., Ltd. As no space-qualified alternatives were available on the market, temperature aging tests of PIN-FET photodetectors, SLD optical source, and MIOC were carried out according to the recommendation of Bell Core reliability standards 'GR-468-Core Generic Reliability Assurance Requirement for Optoelectronic Devices Used in Telecommunications Equipment' [17]. The experiment results show that the mean time between failures (MTBF) of PIN-FET photodetectors should not be less than 324566h, whereas that of the SLD optical source is 150697.6h. As for the MIOC, the test gives MTBF of 1.8×10^7h. The MTBF of the coupler is 4.1×10^6h based on reliability tests [18]. As for FPGA, the results given by the reliability report of Microsemi FPGA and Soc Products are directly used in this work, which reported the MTBF of 0.22 UMC CMOS FPGA is 5.74×10^8h. Additionally, it is supposed that all the electronic components are aero-space and military-grade and the failure rate can be taken less than 10^{-8}h according to the U.S. military standard, Chinese national military standard and other relevant standards [19].

B. Reliability Modeling of Fiber Coils in Space-Borne IFOGs

The largest effect of irradiation on the IFOG is the attenuation in the optical fiber coils, and the long-term radiation exposure in space environment will cause the increase of fiber loss and the reduction of IFOG sensitivity [15]. Once the fiber loss exceeds the specific threshold, the IFOG fails. Most of the literature on the optical fiber is about the modeling of RIA, and the reliability of optical fiber coils used for space application is of limitation. In order to close the gap, a reliability model of optical fiber coils used in the space environment is derived as follows.

The RIA in a fiber obeys a power law of radiation dose d and radiation dose rate r:

$$A = qr^b d^f \tag{2}$$

where q, b and f are constants. For a fixed orbital altitude, the radiation dose rate r is constant and the radiation dose d can be expressed as $d = rt$, where t is the exposure time. So, A can be also expressed as $A = \mu t^f$, $\mu = qr^{b+f}$.

The fiber attenuation $\{A_w(t), t \geq 0\}$ is a continuous-time stochastic process, the Wiener process with drift coefficient μ and infinitesimal variance σ^2 can be employed to model it:

$$A_w(t) = \mu t^f + \sigma B(t) \tag{3}$$

in which μt^f describes the time-dependent change of fiber attenuation, $\sigma B(t)$ indicates the random measurement error. $B(t)$ is a standard Brownian motion and obeys $B(t) \sim N(0, t)$, σ is the diffusion coefficient. Then, the reliability of an optical fiber coil can be derived as:

$$R_w(t) = \Phi(\frac{A_T - \mu t^f}{\sigma\sqrt{t}}) - \exp(\frac{2\mu t^f A_T}{\sigma^2 t})\Phi(-\frac{A_T + \mu t^f}{\sigma\sqrt{t}}) \tag{4}$$

where A_T is the attenuation threshold of the optical fiber coil in dB/km and can be obtained by:

$$A_T = -(10 \lg P_T/P_0 + A_c)/L \tag{5}$$

P_0 is the source power coupled to the optical circuit, L is the length of the fiber coil and Ac includes all the other optical circuit losses due to the fiber couplers, polarizer, MIOC and their pigtails and splices. P_T is the degradation threshold of the optical power, which could be computed by (6), as shown at the bottom of the next page.

RWC is one of the most important performance parameters of IFOGs, which determines the fundamental noise level and the sensitivity of IFOGs. According to the space mission requirements, the degradation threshold of RWC denoted as RWC_T can be fixed. In (6), λ is the operational wavelength, D denotes the diameter of the fiber coil, c represents the speed of light, η means the detector responsivity, e expresses the electron charge, ϕ_b indicates the square wave modulation phase, $\Delta\nu$ is the source spectral bandwidth in the frequency domain, I_d indicates the detector dark current, k denotes the Boltzmann constant, T expresses the temperature and R_d represents the detector load resistance.

Assuming that $\Delta A_w(t_j)$ is an independent increment and its probability density function is

$$\begin{aligned} f&\left(\Delta A_w(t_j); \mu, \sigma, f\right) \\ &= \frac{1}{\sqrt{2\pi\sigma^2\Delta\tau(t_j, f)}} \\ &\quad \times \exp\left\{-\frac{\left[\Delta A_w(t_j) - \mu\Delta\tau(t_j, f)\right]^2}{2\sigma^2\Delta\tau(t_j, f)}\right\} \end{aligned} \tag{7}$$

Then, the corresponding maximum Logarithmic likelihood function can be expressed as:

$$\begin{aligned} \ln L \propto -\frac{1}{2}\sum_{j=1}^{M}&\left\{\left[\ln(2\pi\Delta\tau(t_j, f)) + \ln(\sigma^2)\right]\right. \\ &\left. + \frac{\left[\Delta A_w(t_j) - \mu\Delta\tau(t_j, f)\right]^2}{\sigma^2\Delta\tau(t_j, f)}\right\} \end{aligned} \tag{8}$$

in which the parameters μ, σ^2 can be estimated by:

$$\begin{cases} \overset{\wedge}{\mu} = \sum_{j=1}^{M}\omega_j \Big/ \sum_{j=1}^{M} t_j^f \\ \overset{\wedge}{\sigma}{}^2 = \frac{1}{M-1}\sum_{j=1}^{M}\frac{1}{t_j^f - t_{j-1}^f}(\omega_j - \overset{\wedge}{\mu}\cdot(t_j^f - t_{j-1}^f))^2 \\ j = 1, 2, \ldots, M \end{cases} \tag{9}$$

where ω_j indicates the fiber attenuation data obtained by fiber 60Co-gamma radiation experiments and the details about the irradiation test refer to the relevant literature [20].

C. Reliability Analysis of Multiple-Axis Redundant Configurations for Space-Borne IFOGs

Considering no less than m redundant identical single-axis IFOGs in parallel with same reliability R is abled, the reliability R_N of this redundant configuration can be expressed as:

$$R_N(m, n, R) = \sum_{i=m}^{n} C_n^i R^i (1-R)^{n-i} \tag{10}$$

where $C_n^i = \frac{n!}{i!(n-i)!}$.

Based on the previous studies mentioned in the Introduction, we can obtain the three single-axis configuration, the four single-axis configuration and the six single-axis configuration by duplicating a single-axis IFOG to multiple-axis IFOGs. Considering the case that at least three orthogonal axes of the space-borne IFOG system must be effective during the mission, the reliability of the three single-axis configuration is given by:

$$R_3 = R^3 \tag{11}$$

The reliability of the four single-axis configuration, instead, is:

$$R_4 = \sum_{i=3}^{4} C_4^i R^i (1-R)^{4-i} = 4R^3 - 3R^4 \tag{12}$$

Moreover, the reliability of the six single-axis configuration is:

$$R_6 = \sum_{i=3}^{6} C_6^i R^i (1-R)^{6-i} = 20R^3 - 45R^4 + 36R^5 - 10R^6 \tag{13}$$

D. Calculation Results

The parameters of an experimental IFOG are shown in TABLE III. Based on (6), it can be calculated that the initial RWC $(t = 0)$ of the experimental IFOG is 0.0179 and the bias drift is 0.1°/h (100s). According to the requirements of several space missions, twenty times of magnitude reduction of IFOG accuracy is regarded as IFOG failure, so the RWC_T and A_T can be calculated as 0.342 and 85dB/km respectively. Assuming that the total radiation dose of IFOGs in the space environment for a whole year is 5Krad, whereas half-attenuation of aluminum and lead shielding have been taken into account, then the corresponding radiation dose rate can be calculated as 1.586×10^4 rad/s. On the basis of (6) and the irradiation test data of PM fiber given in [20], the RIA in the fiber can be derived as $A = 7.867 \times 10^7 t^{0.883}$, $\overset{\wedge}{\mu} = 7.867 \times 10^7$, $\overset{\wedge}{\sigma} = 0.0017$.

By substituting the failure rates from TABLE II into the formulas (11)-(13), the reliability of the three single-axis IFOG configuration, four single-axis redundant IFOG configuration and six single-axis redundant IFOG configuration can be obtained as shown in Fig. 3. Furthermore, the estimated values of IFOG modules' weight are listed in TABLE IV.

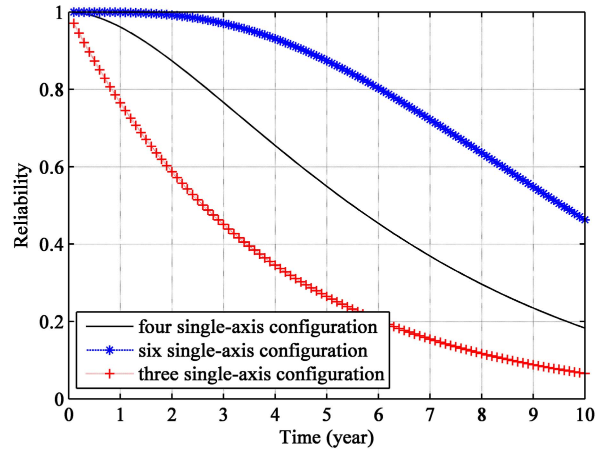

Fig. 3. Reliability of six single-axis redundant configuration (asterisks), four single-axis redundant configuration (straight line) and three single-axis configuration (crosses).

The reliability and the usages of IFOG modules for multi-axis redundant configurations are listed in TABLE V. Compared with the three single-axis configuration, the reliability of the four single-axis configuration improves 25.3% in the first year, 69.7% in the third year and 107% in the fifth year, whereas the usage and the estimated weight of modules grow by 33.3%. The reliability of the six single-axis configuration increases 30.2% in the first year, 114.6% in the third year and 228% in the fifth year, whereas the usage of modules and the estimated weight will double. Based on the calculated results, the redundance of IFOG modules is a feasible way to improve the reliability of the IFOG system. However, the conventional redundance method is not suitable for space application due to the great increase in weight and volume.

IV. Configuration Optimization for Space-Borne IFOGs

A. Configuration Reliability Optimization Method

Based on the reliability analysis and the calculation results of the traditional multi-axis redundant configurations mentioned in the above paragraphs, a configuration optimization method is developed to improve the reliability of space-borne IFOGs with limiting the total weight and the usage of IFOG modules. The IFOG configuration optimization is a criterion to satisfy the requirements of light weight and high reliability for space-borne miniature IFOGs.

We use $R_{G1}, R_{G2}, R_{G3}, R_{G4}$ to represent the reliability of the four basic blocks, $W_{G1}, W_{G2}, W_{G3}, W_{G4}$ indicate their weight, and $N_{G1}, N_{G2}, N_{G3}, N_{G4}$ are their usage. The total weight W_x of an IFOG configuration can be expressed as:

$$W_x = N_{G1}W_{G1} + N_{G2}W_{G2} + N_{G3}W_{G3} + N_{G4}W_{G4} \tag{14}$$

$$RWC_T = \frac{\lambda c}{2\pi LD}\sqrt{\frac{2e(1+\cos\phi_b)^2}{\eta P_T(\sin\phi_b)^2} + \frac{(1+\cos\phi_b)^2}{\Delta\upsilon(\sin\phi_b)^2} + \frac{2eI_d}{(\eta P_T\sin\phi_b)^2} + \frac{4kT}{R_d(\eta P_T\sin\phi_b)^2}} \tag{6}$$

TABLE III
PARAMETERS OF EXPERIMENTAL IFOGS

Symbol	Parameter	Value
P_0	Optical source power	1 mw
L	Length of fiber coil	300 m
Ac	Other optical circuit loss	24.2 dB
λ	Operational wavelength	1310 nm
D	Diameter of fiber coil	35 mm
c	Light speed	3.0×10^{8} m/s
η	Detector responsivity	0.9 A/W
e	Electron charge	1.6×10^{-19} C
ϕ_b	Modulation phase	$\pi/2$
$\Delta\nu$	Source spectral bandwidth in the frequency	6×10^{12} Hz
I_d	Detector dark current	7 nA
k	Boltzmann constant	1.38×10^{-23} J/K
T	Temperature	293.15 K
R_d	Detector load resistance	800 kΩ
q	Radiation environmental parameter	0.00492
b	Radiation environmental parameter	0.117
f	Radiation environmental parameter	0.883

TABLE IV
ESTIMATED VALUES OF IFOG MODULES' WEIGHT

Modules	Weight (g)
Sensing coil module	~50
Optical transmitter module	~15
Optical receiver module	~15
Signal processing module	~10

TABLE V
THE RELIABILITY AND THE USAGES OF IFOG MODULES FOR MULTI-AXIS REDUNDANT CONFIGURATIONS

IFOG Configurations	R_{3G}	R_{4G}	R_{6G}
Optical transmitter module	3	4	6
Optical receiver module	3	4	6
Sensing coil module	3	4	6
Signal processing module	3	4	6
Usage of modules	12	16	24
Estimated weight (g)	~270	~360	~540
Reliability(1 year)	0.765	0.961	0.999
Reliability (3 years)	0.451	0.766	0.971
Reliability (5 years)	0.265	0.549	0.874

The corresponding reliability of the IFOG configuration is denoted as R_x, so the configuration reliability optimization task can be simplified as finding the maximum R_x for a given weight range $\left(0,\ W_x\right)$.

The restrictions for configurations optimization based on the IFOG operating principles and structure characteristics are listed as follows.

Restriction 1: At least one optical transmitter module is needed, ie., $N_{G2} \geq 1$.

Restriction 2: At least three orthogonal sensing coil modules must be effective since three-dimensional attitude is needed for space-borne IFOGs, ie., $N_{G1} \geq 3$.

Restriction 3: The time-division multiplexing modulation method is not considered in our cases, meaning each sensing coil module should correspond to at least one optical receiving module, ie., $N_{G3} \geq N_{G1}$.

Restriction 4: Similar to 3, each receiving module has at least one signal processing module to process the detected signal, ie., $N_{G4} \geq N_{G3}$.

Restriction 5: As for the cases $N_{G1} = N_{G2} = N_{G3} = N_{G4}$, we considered them as multi-axis redundant configurations.

Restriction 6: $N_{G3} = mN_{G1}, N_{G4} = nN_{G3}, m, n = 1, 2, 3 \ldots$.

On the basis of these restrictions and (10), we propose an algorithm to achieve configuration reliability optimization, the steps are summarized as follows.

1) *Inputs:* R_{G1}, R_{G2}, R_{G3}, R_{G4}, N_{G1max}, N_{G2max}, N_{G3max}, N_{G4max},
W_{G1}, W_{G2}, W_{G3}, W_{G4}.
2) *for p=1:* N_{G2max}, *j=1:* N_{G1max}, *m=1:* N_{G3max}, *r=1:* N_{G4max}, *do*
3) *if* $N_{G2}(p)== N_{G1}(j)== N_{G3}(m)== N_{G4}(r)$, *then*
$R_x = R_N(N_{G4}(r),3,\ R_{G1}R_{G2}R_{G3}R_{G4})$
4) *else if* N_{G2}(p)== N_{G1}(j)== N_{G3}(m)!= N_{G4}(r)
&& rem $(N_{G4}(r)/N_{G3}(m),1)$*=0, then*
$R_x = R_N(N_{G3}(m),3,\ R_{G1}R_{G2}R_{G3}) * R_N^3(N_{G4}(r)/N_{G3}(m),1,\ R_{G4})$
5) *else if* $N_{G2}(p)== N_{G1}(j)\ != N_{G3}(m) == N_{G4}(r)$
&& rem$(N_{G3}(m)/N_{G1}(j),1)$*, then*
$R_x = R_N(N_{G1}(j),3,\ R_{G1}R_{G2}) * R_N^3(N_{G4}(r)/N_{G1}(j),1,\ R_{G3}R_{G4})$
6) *else if* $N_{G2}(p)== N_{G1}(j)\ != N_{G3}(m)\ != N_{G4}(r)$
&& rem$(N_{G3}(m)/N_{G1}(j),1)$ *&& rem*$(N_{G4}(r)/N_{G3}(m),1)$ *==0*
then $R_x = R_N(N_{G1}(j),3,\ R_{G1}R_{G2}) * R_N^3(N_{G3}(m)/N_{G1}(j),R_{G3})$
$* R_N^3(N_{G4}(r)/N_{G3}(m),1,R_{G4})$
7) *else if* $N_{G2}(p)!= N_{G1}(j) == N_{G3}(m) == N_{G4}(r)$*, then*
$R_x = R_N(N_{G2}(p),1,\ R_{G2}) * R_N(N_{G4}(r),3,\ R_{G1}R_{G3}R_{G4})$
8) *else if* $N_{G2}(p)!= N_{G1}(j) == N_{G3}(m)\ != N_{G4}(r)$*, then*
$R_x = R_N(N_{G2}(p),1,\ R_{G2}) * R_N(N_{G3}(m),3,\ R_{G1}R_{G3})$
$* R_N^3(N_{G4}(r)/N_{G3}(m),1,R_{G4})$
9) *else if* $N_{G2}(p)!= N_{G1}(j)\ != N_{G3}(m) == N_{G4}(r)$ *&& rem*$(N_{G3}(m)/N_{G1}(j),1)$*, then*
$R_x = R_N(N_{G2}(p),1,\ R_{G2}) * R_N(N_{G1}(m),3,\ R_{G1})$
$* R_N^3(N_{G4}(r)/N_{G1}(j),1,\ R_{G3}R_{G4})$
10) *else if* $N_{G2}(p)!= N_{G1}(j)\ != N_{G3}(m)\ != N_{G4}(r)$ *&& rem*$(N_{G4}(r)/N_{G3}(m),1)$ *==0 && rem*$(N_{G3}(m)/N_{G1}(j),1)$*,*
then $R_x = R_N(N_{G2}(p),1,\ R_{G2}) * R_N(N_{G1}(m),3,\ R_{G1})$
$* R_N^3(N_{G3}(m)/N_{G1}(j),1,\ R_{G3}) * R_N^3(N_{G4}(r)/N_{G3}(m),1,\ R_{G4})$
end if
11) $W_x = N_{G1}(j)*W_{G1}+N_{G2}(p)*W_{G2}+N_{G3}(m)*W_{G3}+N_{G4}(r)*W_{G4}$, *end for*
12) *Outputs:* N_{G1}, N_{G2}, N_{G3}, N_{G4}, R_x, W_x.

The flow chart of the configuration reliability optimization method is illustrated in Fig. 4.

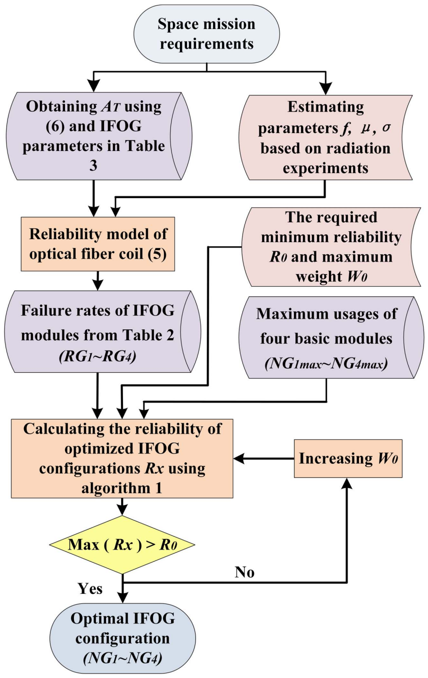

Fig. 4. Flow chart of configuration reliability optimization method for space-borne IFOGs.

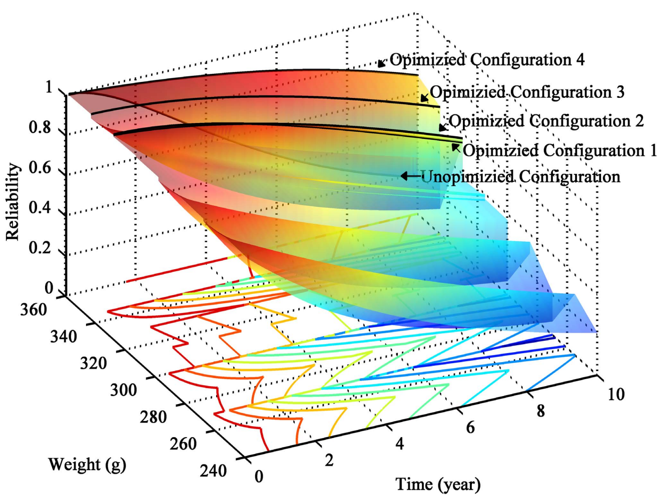

Fig. 5. Reliability surface of IFOG configurations.

B. Optimization Results

In order to demonstrate this proposed configuration optimization method, we optimized the four single-axis IFOG configuration based on our experimental IFOG's parameters. The maximum weight W_0 (the total weight of the four single-axis redundant configuration) and the required minimum reliability R_0 are set to be 360g and at least 0.8 in the fifth year respectively.

As presented in Fig. 5, all the IFOG configurations that satisfy the weight requirement are considered and the reliability surface of each configurations is provided. There are four optimized configurations and Fig. 6 plots the reliability comparison between the optimized configurations and the unoptimized configuration. Obviously, these four optimized configurations not only guarantee light weight and small volume but also offer the advantage of high reliability for space-borne IFOGs. The detailed data are listed in TABLE VI.

The first optimized configuration, denoted as four-axis with source-sharing configuration, has two optical transmitter modules, four optical receiver modules, four signal processing modules and four sensing coil modules. The diagram of this configuration is shown in Fig. 7. Compare to the traditional

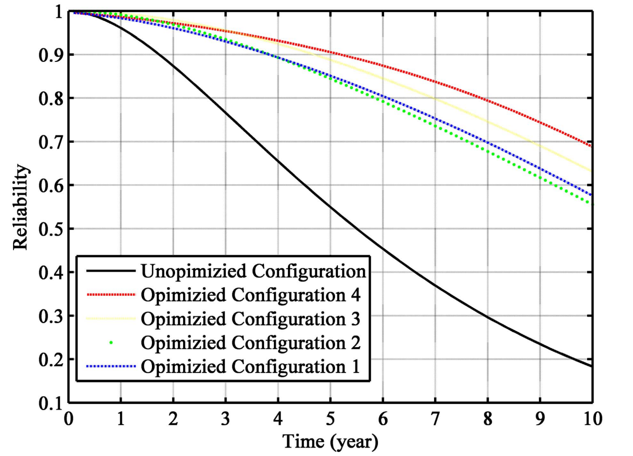

Fig. 6. Comparison between optimized configurations and the four single-axis redundant configuration.

TABLE VI
COMPARSION BETWEEN THE OPTIMIZED CONFIGURATIONS AND THE UNOPTIMIZED CONFIGURATION

Configurations	Optimized 1	Optimized 2	Optimized 3	Optimized 4	Unoptimized
Sensing coil module	4	3	4	3	4
Optical transmitter module	2	2	3	4	4
Optical receiver module	4	6	4	6	4
Signal processing module	4	6	4	6	4
Weight (g)	330	330	345	360	360
The usage of modules	14	17	15	19	16
Reliability (1 year)	0.991	0.983	0.994	0.987	0.961
Reliability (3 years)	0.935	0.930	0.956	0.954	0.766
Reliability (5 years)	0.845	0.851	0.888	0.906	0.549

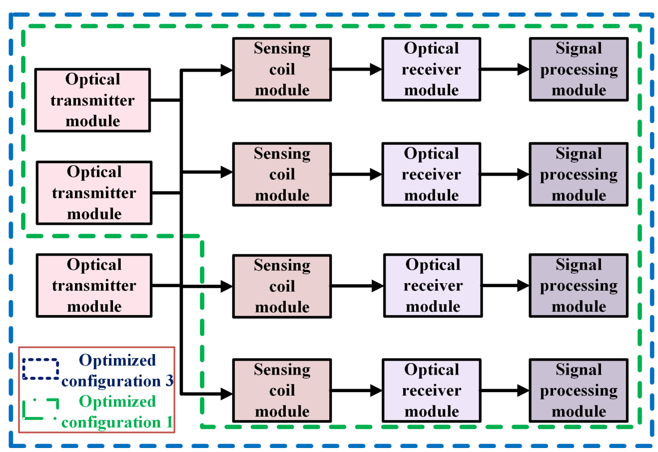

Fig. 7. Configuration diagram of optimized configuration 1 and optimized configuration 3.

TABLE VII
RELIABILITY RESULTS FOR 1000000 SIMULATIONS AND DIFFERENT YEARS USING DIFFERENT METHODS

Configuration	R_{3G}		R_{4G}		R_{6G}		Optimized 1		Optimized 2	
Method	MCM	RAM	MCM	RAM	MCM	RAM	MCM	RAM	MCM	RAM
Reliability (1 year)	0.762	0.765	0.960	0.961	0.999	0.999	0.991	0.992	0.983	0.987
Reliability (3 years)	0.426	0.451	0.742	0.766	0.971	0.963	0.935	0.929	0.930	0.928
Reliability (5 years)	0.252	0.265	0.522	0.549	0.826	0.826	0.817	0.818	0.830	0.831

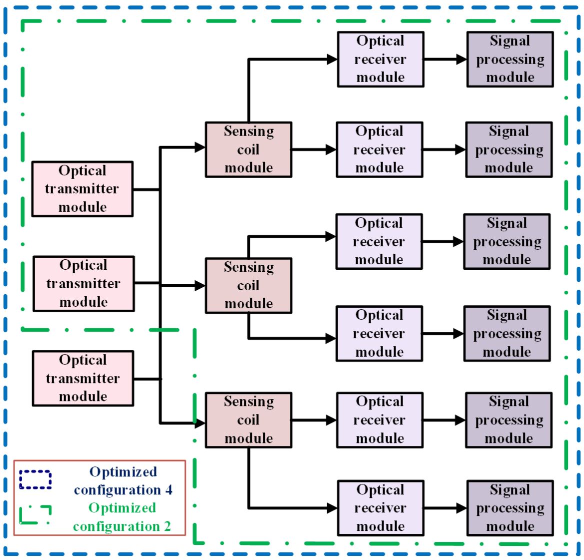

Fig. 8. Configuration diagram of optimized configuration 2 and optimized configuration 4.

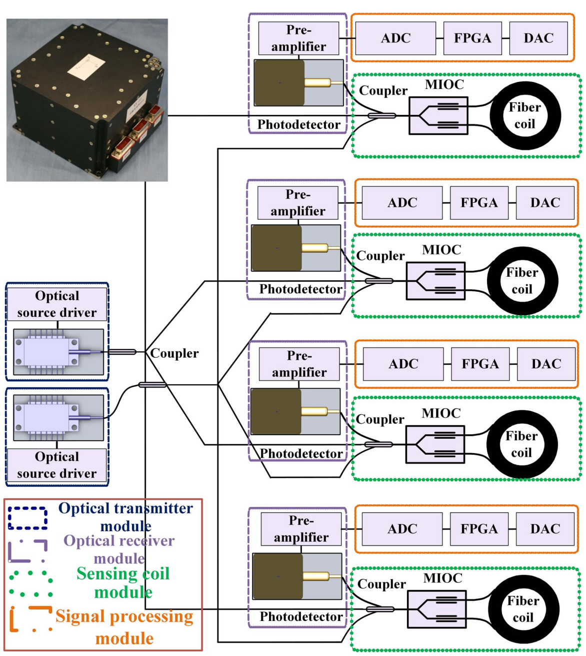

Fig. 9. Schematic diagram of optimized configuration 1.

four single-axis redundant configuration, the reliability raises 3.1% in the first year, 22.1% in the third year and 53.9% in the fifth year, with two fewer modules and the total weight dropped by 8.3%. For the second optimized configuration,

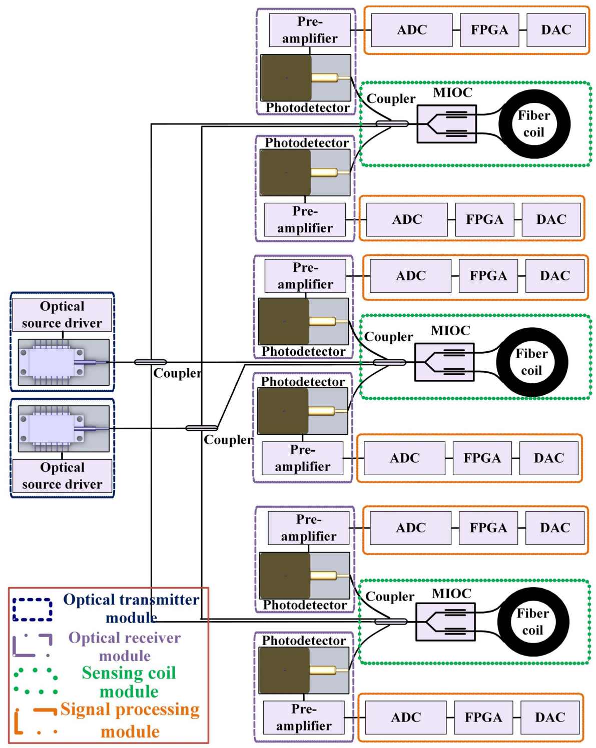

Fig. 10. Schematic diagram of optimized configuration 2.

an optical receiver module for each axis and an optical transmitter are redundant, meanwhile, both two optical transmitter modules are multiplexing. The diagram of this configuration is shown in Fig. 8. Compared with the traditional four single-axis configuration, the reliability greatly improves 2.3% in the first year, 21.4% in the third year and 55.0% in the fifth year, with one additional module and the total weight also decreased by 8.3%. As for the third case in the optimized results, it adds an optical transmitter module on the basis of optimized configuraiton 1. The diagram of this configuration is plotted in Fig. 7 with dashed blue lines. Compared to the traditional four single-axis redundant configuration, the reliability raises 3.4% in the first year, 24.8% in the third year and 61.7% in the fifth year, with one less module and the total weight dropped by 4.2%. Similarly, the fourth optimized configuration also adds an optical transmitter module based on optimized configuration 2. Compared with the traditional four single-axis configuration, the reliability improves 2.7% in the first year, 24.5% in the third year and 65.0% in the fifth year, with three additional modules and the total weight be the same. Correspondingly, the configuration diagram is plotted in Fig. 8 with dashed blue lines.

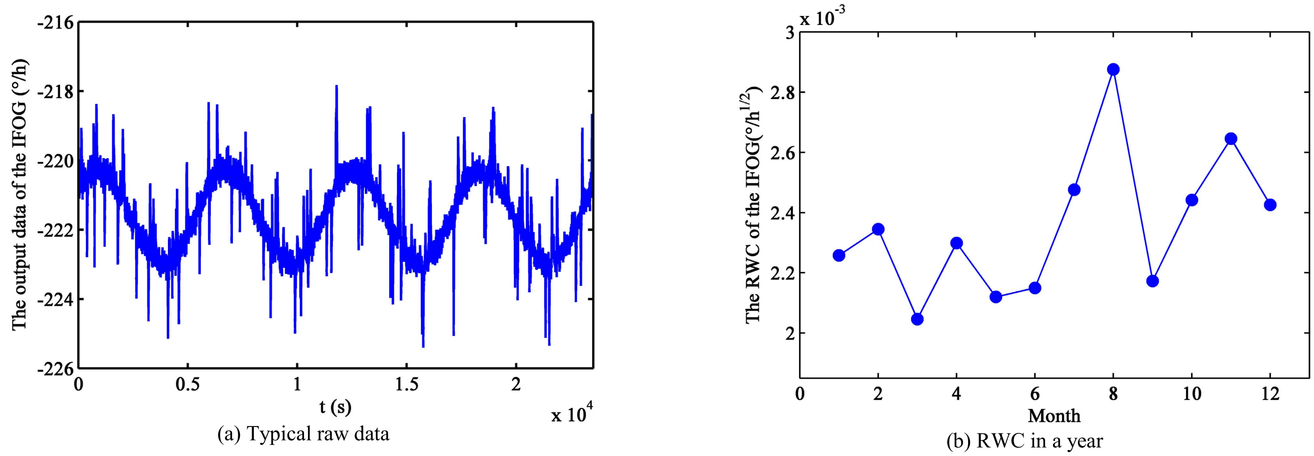

Fig. 11. In-orbit test results of Y-axis of SF4 × 135.

Further analysis is carried out according to other application conditions. Although the reliability of optimized configuration 4 is 0.4%-2.0% higher during 1-5 years, the usage of IFOG modules is apparently larger than the other three optimized configurations, which indicates that the volume of this configuration is bigger since the volume of IFOGs is mainly determined by the usage and architecture of the modules. Moreover, compared to optimized configuration 2, optimized configuration 3 has 1.1%-4.2% higher reliability during 1-5 years but adds one more optical transmitter module, which means higher power consumption in this configuration. To sum up, optimized configuration 1 and optimized configuration 2 are optimal configurations.

V. Validation of Configuration Optimization Method

A. Validation Based on Monte Carlo Method

In the light of probability theory, the MCM was proposed for applications in many fields due to high precision for complex failure mode [21]. To validate the simulation results in TABLE V and TABLE VI, the MCM was adopted to analyze the reliability of IFOG configurations with the experimental IFOG parameters in TABLE III and the failure rates of IFOG components in TABLE II by the same computer configuration. TABLE VII lists the comparison results for 100 0000 simulations and different years using the MCM and the reliability assessment model (RAM) derived in Section II, where the MCM is considered as the reference. From the comparative results, one can see that the deviation between the two methods is very small, and the maximum deviation is only 5.2%, which demonstrate that the reliability simulation results of IFOG configurations in TABLE V and TABLE VI is feasible, and correspondingly, the established configuration optimization method is effective to improve the reliability of IFOGs without increasing weight and volume.

B. Ground-Based Tests

The optimized configuration 1 has been implemented in our typical IFOG product SF4 × 135. Its configuration scheme is shown in Fig. 9 and its main performances are shown in TABLE VIII. To verify the reliability of this optimal configuration, the ground-based environmental acceleration tests were carried out. The test items and their brief conditions are listed in TABLE IX. There are ten SF4 × 135 in the acceleration tests, whose performances are monitored in real-time. The bias drift, the RWC and the scale factor accuracy (SFA) of all these ten SF4 × 135 before and after the ground-based acceleration tests are compared, as presented in TABLE X. The comparison results illustrate that there is no obvious change for all these ten IFOGs' performances before and after the acceleration tests and all satisfy the index requirements, which support that the high reliability of the optimized configuration 1. Besides, the configuration scheme of optimized configuration 2 is plotted in Fig. 10 and its prototype has been developed.

TABLE VIII
Performance of IFOG SF4 × 135.

Parameters	Index	Parameters	Index
Bias drift	⩽0.02° /h	Reliability (3 years)	>0.9
RWC	⩽0.005° /√h	Weight	⩽1.7kg
Scale factor accuracy	⩽50ppm[a]		

[a]ppm=10^{-6}

TABLE IX
Ground-Based Environmental Acceleration Test Items and Their Conditions for IFOG SF4 × 135

Test items	Test conditions
Thermal cycling	–25°C/+50°C, 250h
Thermal vacuum cycling	–25°C/+50°C, 50h, ≤1.3×10-3 Pa
Vibration test	20 to 2000Hz, 10G
Impact test	1000 g, 3 directions, non-powered

C. In-Orbit Validation

Additionally, the SF4 × 135 has been actually used in a commercial remote sensing satellite for 3 years and performs

TABLE X
COMPARISON OF IFOG SF4 × 135'S PERFORMANCES BEFORE AND AFTER GROUND-BASED ENVIRONMENTAL ACCELERATION TEST (SOURCE 1, Y-AXIS)

Test items	1	2	3	4	5	6	7	8	9	10
Bias drift before test (°/h)	0.0145	0.0187	0.0143	0.0189	0.018	0.018	0.0198	0.0152	0.0188	0.0166
Bias drift after test (°/h)	0.0149	0.0195	0.0145	0.0191	0.0192	0.0185	0.019	0.0147	0.0155	0.0161
Bias drift before and after variation	0.0004	0.0008	0.0002	0.0002	0.0012	0.0005	-0.0008	-0.0005	-0.0033	-0.0005
RWC before test (°/($h^{1/2}$))	0.0023	0.003	0.0024	0.0012	0.0014	0.0034	0.0032	0.0023	0.0036	0.0024
RWC after test (°/($h^{1/2}$))	0.0024	0.0034	0.0025	0.0017	0.001	0.0035	0.0045	0.0024	0.0033	0.0025
RWC before and after variation	0.0001	0.0004	0.0001	0.0005	-0.0004	0.0001	0.0013	0.0001	-0.0003	0.0001
SFA before test (ppm)	2.98	9.01	12.46	17.6	7.8	12.43	2.44	21.15	13.16	15.6
SFA after test (ppm)	2.31	1.94	8.8	11.9	2.75	26.55	14.27	11.74	49.79	30.12
SFA before and after variation	-0.67	-7.07	-3.66	-5.7	-5.05	14.12	11.83	-9.41	36.63	14.52

satisfactorily, which also validates the high-reliability of the optimized configuration 1. The in-orbit test results of the Y-axis of SF4 × 135 are shown in Fig. 11.

VI. CONCLUSION

A configuration optimization method for improving the reliability of space-borne IFOGs in a limited weight range was proposed and verified in this paper. The effects of the space environment on IFOG components were firstly analyzed. Then, the reliability assessment model for space-borne IFOGs was derived based on the failure characteristics of these components. Next, all the configurations that satisfied the IFOGs' working principles and the requirements of limited weight are considered in the proposed configuration optimization method. From the optimized results, one can see that compared with the conventional four single-axis redundant configuration, the optimized configurations with our experimental IFOG parameters improve the reliability by 2.3%-3.1% in the first year, 21.4%-22.1% in the third year and 53.9%-55.0% in the fifth year, whereas the total weight dropped by 8.3% and the usage of IFOG modules is almost the same. The feasibility of the simulation results and the effectiveness of the reported configuration reliability optimization method are verified by comparing with the MCM. Indeed, the established method has been employed in our space-borne IFOG prototypes and products, whose ground-based accelerated environmental test results and the in-orbit test results also support the high reliability of the optimized configuration. We believe that the configurations optimized using our method could meet the requirements of high reliability and light weight for miniature space-borne IFOGs.

REFERENCES

[1] H. C. Lefevre, *The Fiber-Optic Gyroscope*, 2nd ed. London, U.K.: Artech House, 2014.

[2] X. Zheng and W. Wei, "Reliability study of redundant configuration of IMU fiber optic gyroscope inertial measurement unit," in *Proc. 8th Int. Conf. Rel., Maintainability Saf.*, Chengdu, China, Jul. 2009, pp. 87–90.

[3] H. J. Kim, C. G. Park, and J. W. Song, "Analytic optimal solution of multiple gyros configuration based on FDI/GNC performance," *Electron. Lett.*, vol. 52, no. 19, pp. 1633–1635, Sep. 2016, doi: 10.1049/el.2016.1951.

[4] R. Fontanella *et al.*, "Exploiting low-cost compact sensor configurations performance by redundancy," in *Proc. AIAA Inf. Syst.-AIAA Infotech Aerosp.*, 2018, p. 0194.

[5] G. McVittie and K. Kumar, "Design of a COTS femtosatellite and mission," in *Proc. AIAA SPACE Conf. Expo.*, 2007, p. 6034.

[6] N. Song, W. Cai, J. Song, J. Jin, and C. Wu, "Structure optimization of small-diameter polarization-maintaining photonic crystal fiber for mini coil of spaceborne miniature fiber-optic gyroscope," *Appl. Opt.*, vol. 54, no. 33, p. 9831, Nov. 2015.

[7] N. Song, K. Ma, J. Jin, F. Teng, and W. Cai, "Modeling of thermal phase noise in a solid core photonic crystal fiber-optic gyroscope," *Sensors*, vol. 17, no. 11, p. 2456, 2017.

[8] P. Roberts, F. Couny, and H. Sabert, "Ultimate low loss of hollow-core photonic crystal fibres," *Opt. Express*, vol. 13, no. 1, pp. 236–244, 2005.

[9] X. Xu, Z. Zhang, Z. Zhang, J. Jin, and N. Song, "Method for measurement of fusion-splicing-induced reflection in a photonic bandgap fiber-optical gyro," *Chin. Opt. Lett.*, vol. 13, no. 3, pp. 030601–030604, 2015.

[10] Z. Zhang, X. Xu, N. Song, C. Zhang, and F. Teng, "Reflection induced noise analysis of an air-core photonic bandgap fiber optic gyroscope," in *Proc. Photon. Fiber Technol. (ACOFT, BGPP, NP)*, 2016, Paper ATh2C.7

[11] H. Kanai, K. Hama, M. Akiyama, N. Oka, K. Mitani, and M. Arakaki, "SERVIS project: Application of commercial technology for space," in *Proc. Int. Symp. Space Technol. Sci.*, 2002, vol. 23, no. 2, pp. 2628–2635.

[12] S. Esposito *et al.*, "COTS-based high-performance computing for space applications," *IEEE Trans. Nucl. Sci.*, vol. 62, no. 6, pp. 2687–2694, Dec. 2015.

[13] S. K. Bhattacharyya and A. S. Cheliyan, "Optimization of a subsea production system for cost and reliability using its fault tree model," *Rel. Eng. Syst. Saf.*, vol. 185, pp. 213–219, May 2019.

[14] M. Van Uffelen and P. Jucker, "Radiation resistance of fiberoptic components and predictive models for optical fiber systems in nuclear environments," *IEEE Trans. Nucl. Sci.*, vol. 45, no. 3, pp. 1558–1565, Jun. 1998.

[15] R. H. Boucher *et al.*, "Proton-induced degradation in interferometric fiber optic gyroscopes," *Opt. Eng.*, vol. 35, no. 4, pp. 120–163, Apr. 1996.

[16] H. Gao, A. Wang, G. Bai, C. Wei, and C. Fei, "Substructure-based distributed collaborative probabilistic analysis method for low-cycle fatigue damage assessment of turbine blade–disk," *Aerosp. Sci. Technol.*, vol. 79, pp. 636–646, Aug. 2018.

[17] X. Liu, X. Wang, H. Xiao, and B. Liu, "Reliability analysis on operation of a kind, of all fiber-optic current transformer," in *Proc. China Int. Conf. Electr. Distrib. (CICED)*, Shenzhen, China, Sep. 2014, pp. 1519–1523.

[18] Y. Abe, K. Shikama, S. Asakawa, and S. Yanagi, "20-year reliability test results for SC connector installed on outside plant," in *Proc. Opt. Fiber Commun. Conf.*, San Diego, CA, USA, 2018, pp. 1–3.

[19] *Department of Defense Standard Practice: Failure Rate Sampling Plans and Procedures*, Standard MIL-STD-690, 2005.

[20] J. Jin and S. Lin, "Parameters optimization of interferometric fiber optic gyroscope for improvement of random walk coefficient degradation in space radiation environment," *Opt. Lasers Eng.*, vol. 50, no. 11, pp. 1542–1547, Nov. 2012.

[21] H. Gao, C. Fei, G. Bai, and L. Ding, "Reliability-based low-cycle fatigue damage analysis for turbine blade with thermo-structural interaction," *Aerosp. Sci. Technol.*, vol. 49, pp. 289–300, Feb. 2016.

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